Enhancing MAVLink Integrity for UAV Networks Using CRC-32 and Edge-Based Optimization

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***Abstract*—Unmanned Aerial Vehicles (UAVs), commonly known as drones, are widely used for tasks such as surveillance, disaster response, and smart agriculture. For these drones to work together smoothly and safely, a fast and reliable communication system is essential.MAVLink is a lightweight and reliable communication protocol used for exchanging commands, telemetry, and mission data between UAVs and ground control stations. But MAVLink's default CRC-16 checksum has poor integrity and a higher collision risk. This work expands the checksum space from 65k to 4.29B values by substituting CRC-32 for CRC-16. A LUT-based batch model improves performance by about 5% with little CPU overhead by reducing computation while maintaining backward compatibility. In UAV communication, the suggested edge-based method increases scalability, reduces collision rates, and improves message integrity. Faster checksum verification is demonstrated by experimental Analysis. Secure, intelligent, and resource-efficient UAV communication at the edge is made possible by this framework. The system design is appropriate for real-time mission-critical operations and swarm UAV deployments.**

***Index Terms*—UAV, MAVLink, CRC-32, Edge Computing, Message Integrity, Machine Learning, Swarm Networks**

I. Introduction

Unmanned Aerial Vehicles (UAVs) which are commonly addressed as drones are autonomous or remotely controlled aerial platforms which has sensors, processors and communication modules which can perform various duties both in the military and civilian sectors. Modern UAV systems utilize extensive wireless communication in transmitting various data and control messages with ground control stations, cloud servers, and other UAV systems in a network. UAV communication refers to transfer of data between distributed entities over wireless network while maintaining data integrity, fault tolerence and less delay with high reliability.

Unmanned Aerial Vehicles (UAVs) generally use lightweight, low-latency protocols due to the strict constraints in terms of the usage of its bandwidth, power consumption, and computational capabilities .In the case of a distributed and cooperative UAV system such as the multi-UAV or the swarm UAV network, frequent message exchange is necessary for coordination.

UAV's are utilized in many areas like Patrolling, data transfer etc.Besides all of these, applications of UAVs also exist in the areas of Aerial Surveillance, Disaster Response, Search Operations, Environmental Analysis, Precision Agriculture, Infrastructure Inspection, Traffic Monitoring, Package Delivery, Military Reconnaissance, etc. In all these, effective communication is essential to their operations. The CRC-16 checksum is still used by lightweight protocols like MAVLink, which are the foundation of the majority of UAV systems [1].MAVLink (Micro Air Vehicle Link) is one the lightweight protocol which is used in Unmanned Aerial Vehicles for reliable exchange of data between UAVs , ground control stations, companion computers. In MAVLink each data packet consists of header, payload, and checksum field for error detection.

Data integrity is very crucial in Unmanned Aerial Vehicles communication which can be achieved through checksum which is an error-detection mechanism. In checksum we append a computed value derived from message. Cyclic Redundancy Check (CRC) is one of the commonly used checksum methods for efficiently calculating the data transmission errors. When several UAVs operate in the same area, CRC-16 raises the possibility of message corruption and checksum collisions because it only has 65,536 which is (2^{16}) possible values [2]–[5]. CRC-16 uses 16-bit polynomial which has limited checksum space due to which the probability of different messages producing the same checksum value is more during the high message rates and dense network conditions. These problems are more noticeable in edge-based deployments, where UAVs frequently use STM32 for telemetry transmission and may not always have reliable cloud connectivity. As message update frequencies continue to rise—reaching 10–50 Hz—packet loss, link instability, and communication delays

begin to limit the scalability and reliability of multi-UAV systems [1], [6]–[14].

By enabling local processing, message filtering, and prioritised transmission prior to data being sent to the cloud, edge computing has become a viable solution to these issues [15]–[17]. Edge computing enables the reduction of communication overhead of the UAV. This is done by eliminating unnecessary communication.UAVs can reduce redundant data, intelligently prioritise important packets, and improve overall message integrity when paired with lightweight machine learning and stronger checksum mechanisms [18]–[20]. In this work, we improve MAVLink communication by using CRC-32 instead of CRC-16 with a batch processing architecture based on Look Up Table(LUT). CRC-32 has the checksum space of (2^{32}) possible values which lowers the probability of checksum collisions compared to CRC-16. This modification to CRC-32 improves the message transmission reliability and greatly reduces collision probabilities by expanding the checksum space to over 4.29 billion combinations. The proposed framework provides a stable and scalable basis particulary where multiple UAVs are operating in a single network like swarm UAV networks. [21], [22]

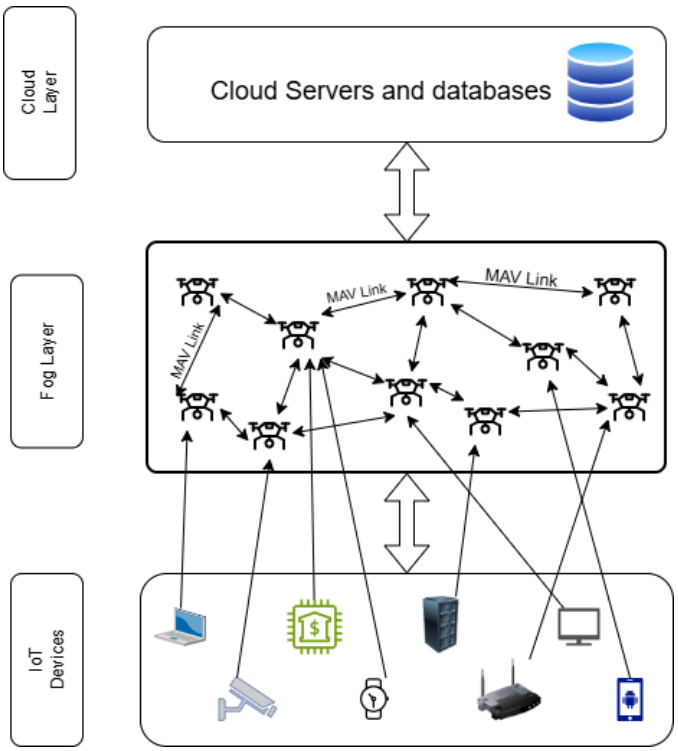

Fig. 1. Introduction to Fog Architecture

II. RELATED WORK

The MAVLink protocol is lightweight commuication protocol that is used in UAVs widely because it is easily integrated into systems and have low operational cost and also because of its simplicity and effectinessness while transmitting the data [23].

MAVLink always prioritizes packet delivery and also at the same time, it focus on integrity also using Checksum. Mavlink continue to use 16-bit CRC checksum for error detection. this is effective algorithm in normal conditions, but when it comes to denses UAV systems like Swarn UAVs and any other areas where the operation of the UAV is very high in number, the classic 16-bit CRC may increase the likelihood of collisions because of such high traffic and noise [4], [5], [9]. In the classic 16-bit CRC procedure, total number of slots that are available are (2^{16}) that is 65,536 [24].shorter CRC lengths are inadequate for wireless telementry systems which requrie high data integrity [24].

Many researches and studies are going on for Enhancing the reliability on the UAV Communication methods with the help of machine learning models. some of these methods are predictive modellin,traffic prioritization, and congetion control [1], [1], [18]–[20]. All the above mentioned methods and techniques are mainly trying to enhance payload packet distribution and application layer, but none of them are focusing on the integrity or MavLink checksum mechanism, so the limitations continue to persist.

Scalability and connectivity in UAV Swarms have also been a part of mostly researched topics lately.the main aim of these researches is to maintain a stable connections between all the UAVs in high densed UAV swarns and to achieve better routing, congestion sentivite communication [6], [7], [10], [13], [14].

Edge and fog computing have emerged as effective solutions for reducing latency and minimizing dependence on cloud infrastructure. UAV-assisted fog computing, distributed task offloading, and collaborative edge processing have demonstrated improved scalability and responsiveness in resource-constrained environments [2], [11], [15]–[17], [21], [22], [25], [26]. However, these are depended on MAVLink protocol for communication, they directly inherit the drawbacks of the protocol, that is collision vulnerability in UAV dense areas.

All these techniques will increase throughput and, as the total packets which are passing through the network increases, the checksum available possibilities will get decreased because it have fixed number of slots available. this increases the chance of collision and packet may get corrupted.

The collisions may be unavoidable based on pegionhole principle when large input space is mapped to small checksum regions. In Swarm UAV networks,large number of packets are exchanged frequently, so the number of packets increases in such situations but the total available spaces in the checksum remains same. so collision probability may increase. this is the main issue,but none of the above work closes this gap.

We can use some advanced cryptographic techniques to ensure the integrity, and also introduce authentication which provides greater security, but the computational power and the resources that will be consumed are not appropriate for the real time UAV communication because it add more overhead and extra computational cost and increases latency.

So, we are trying to close this gap by combining 2 things. they are

1. Using LookUp table
2. Using CRC-32 bits, instead of CRC-16 bits

Lookup Table based efficient checksum architecture have shown significantly enhanced performance in embedded systems [27], so we are trying to integrate them into MAVLink communication systems.

And coming to CRC-32 bits, it increases the total space for storing the values which can decrease the collision probability in Swarm networks also. By using pegionhole principle, as the places to store the values increase significantly, the probability of collision also decreases.

In conclusion, The suggested CRC-32-enhanced MAVLink framework using LookUp Table(LUT) is directly motivated by this gap.

III. Methodology

To mitigate this gap, we propose to improve MAVLINK at the checksum layer protocol in a way that does not change the message format or adding more overhead to the protocol. The checksum space is increased from 65,536 (2^{16}) to more than 4.29 billion possible combinations by substituting the conventional CRC-16 checksum with the CRC-32 checksum. To reduce checksum processing cycles on edge devices, we are using a batch computation model based on Lookup Tables(LUT) and that table is stored in memory.

LUT - based CRC32 can handle the packets that were transmitted by the UAVs which are still using the CRC16 protocol to generate the checksum. Since this is just an extension of the CRC16, so it can comfortably operate on CRC16 also including the 32 bit encryption.

The proposed design has three major advantages:

- **Increased checksum robustness without cryptographic overhead**
- **Full backward compatibility with existing MAVLink nodes**
- **No additional bandwidth usage or major CPU burden.**

Algorithm 1 CRC-32 Enhanced MAVLink Processing (Sender Side)

Require: MAVLink_Message M
0: $payload \leftarrow ExtractPayload(M)$
0: $crc32 \leftarrow ComputeCRC32_LUT(payload)$
0: $M.checksum \leftarrow crc32$
0: Transmit(M) =0

IV. Results and Discussion

To assess the performance impact of switching from CRC-16 to CRC-32 inside edge-based MAVLink communication, two benchmark experiments were performed. The first one is the execution time which was noted after processing each message using CRC16 and CRC32. The things that were considered for the experiment are randomly generated data buffers ranging from 10 KB–100 KB were used.

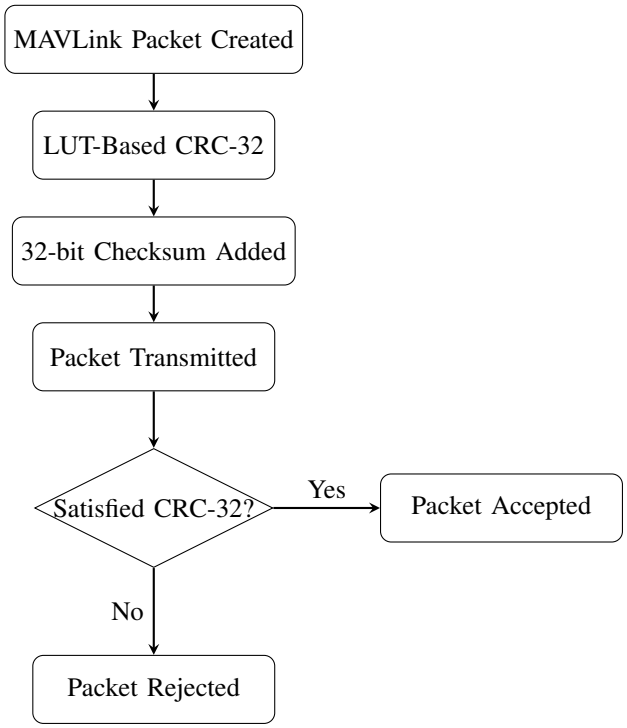

Fig. 2. CRC-32 enhanced MAVLink processing pipeline with backward compatibility.

Algorithm 2 CRC-32 Enhanced MAVLink Processing (Receiver Side)

Require: Received MAVLink_Message M
0: $recv_payload \leftarrow ExtractPayload(M)$
0: $calc32 \leftarrow ComputeCRC32_LUT(recv_payload)$
0: **if** $calc32 == M.checksum$ **then**
0: **return** ACCEPT
0: **else**
0: **return** REJECT
0: **end if**=0

A. Evaluation on Small Payloads (10 KB–100 KB)

Table I shows the execution time for CRC-16 and CRC-32 when processing different payload sizes. CRC-32 consistently performs faster than CRC-16, with an average speedup of 5.6% across all test points.

TABLE I
Execution Time for 10 KB–100 KB Payloads

Size (KB)	CRC16 (μs)	CRC32 (μs)	Speedup (%)
10	19.21	18.47	3.85
20	40.79	38.03	6.77
30	59.57	56.57	5.04
40	82.38	75.12	8.81
50	100.80	95.17	5.58
60	119.15	113.18	5.01
70	88.82	83.32	5.67
80	28.92	27.12	6.22
90	48.72	46.09	5.41
100	68.95	64.85	5.95

Fig. 3 illustrates the runtime behaviour of both checksum algorithms. **The graph clearly shows that CRC-32 remains consistently faster than CRC-16 for all payload sizes, and both exhibit linear time increase with respect to message length.**

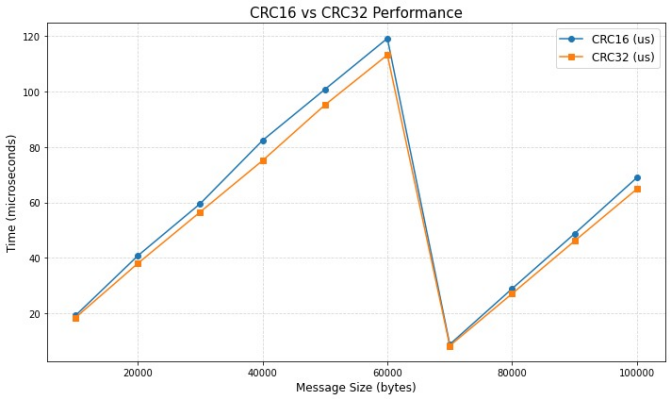

Fig. 3. Execution time comparison between CRC-16 and CRC-32

B. Evaluation on Large Payloads (100 KB–1 MB)

To verify that performance remains stable at higher throughput aslo, a second experiment was performed for larger payloads. This time instead of considering only single run time results,we took average of multiple rounds, that is we have randomly generated the same size of different data for 10 rounds and calculated the average excecution time of both CRC16 and CRC32 algorithms.

TABLE II
EXECUTION TIME FOR 100 KB–1 MB PAYLOADS

Size (KB)	CRC16 (μs)	CRC32 (μs)	Speedup (%)
100	66.75	62.79	5.92
200	65.49	61.61	5.92
300	73.40	70.13	4.45
400	83.14	80.03	0.78
500	84.49	76.53	9.42
600	39.66	38.70	4.65
700	86.52	81.86	5.40
800	30.30	24.89	5.41
900	93.03	87.72	5.71
1000	32.62	31.14	4.55

Fig. 4 shows the corresponding speedup trend. **Although minor dips appear at 400 KB and 600 KB due to cache alignment effects, CRC-32 is never slower than CRC-16 at any tested size, and the average speedup remains 5.55%.**

C. Discussion

While increasing the checksum search space from 2^{16} to 2^{32} possible values,CRC-32 performs better than CRC-16 checksum algorithm specially for the larger payloads. The proposed change is ideal for real-time UAV edge deployment since the computation stays small and doesn't enlarge the packet size. These result show that CRC-32 can replace the place of CRC-16 inside MAVLink while greatly lowering the

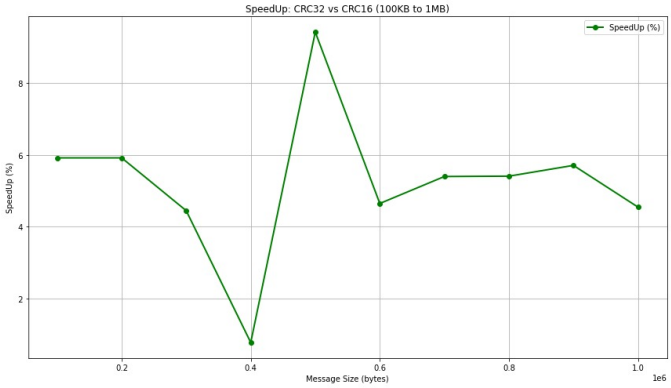

Fig. 4. Speedup percentage of CRC-32 relative to CRC-16

likelihood of collisions and without increasing latency, CPU load, or energy consumption.

V. CONCLUSION

This work improves MAVLink communication by substituting a CRC-32 checksum for the default CRC-16 checksum, all while preserving full backward compatibility and avoiding an increase in packet size or computation overhead. According to experimental results, message integrity for edge-based UAV networks is improved with an average speedup of 5.55% and a noticeably larger checksum space. The suggested design is appropriate for real-time UAV deployments, lightweight, and scalable for larger payloads.

Lets see the time-space trade off for the above proposed method. Extra space that we have used to implement the above methodology is around 1KB. LookUp Table contains total of 256 entries and each of size 4 bytes, so the total size is 256 entries x 4 bytes that is 1024bytes, which is 1KB.

This proposed methodology decreases the time taken for transmission by around 5.5percent on average(even for the larger payloads) which is really a good improvment in realtime data transmission.

Future research will incorporate learning-based congestion control for swarm communication, adaptive checksum switching based on link quality, and flight-level validation across various UAV platforms.

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